

Probabilistic Spatiotemporal Air Quality Forecasting via Physics-Guided Attention and Variational Inference

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1 Introduction

The evolution of society has brought about several irreversible developments, one of the most notable being industrialization. These advancements have had a profound impact on nearly every aspect of life; however, one of the most critical consequences concerns human health and overall well-being, among which air quality has emerged as a major issue. This prominence stems from the excessive emissions generated by industrialization and transportation activities, both of which have become inseparable parts of modern society. Harmful pollutants such as PM2.5, SO, CO, and CO released from these activities significantly degrade air quality. Due to their ability to disperse and circulate easily in the atmosphere, these pollutants expose a large population to hazardous conditions, particularly in densely populated urban areas where both human activity and emission sources are highly concentrated.

A wide range of efforts has been made to mitigate these impacts, ranging from data collection and analysis to the development of mitigation strategies. Among these, air quality analysis plays a critical role by enabling more accurate, evidence-based, and effective decision-making. In particular, air quality forecast with high accuracy is considered an effective way to mitigate detrimental impact of pollutant exposure [6]. Consequently, air quality forecasting has become an important research area in environmental science. Traditional forecasting methods primarily rely on historical observations to predict future values over time. Nevertheless, because air quality exhibits strong spatial dependency, incorporating spatial information can effectively complement temporal patterns. This integration provides richer contextual information, analogous to understanding the surrounding environment in addition to observing how a phenomenon evolves over time. As a result, spatial-temporal forecasting models can better capture the underlying dynamics, leading to more accurate, representative, and realistic predictions.

2 Related Studies

Machine learning (ML) models have been widely employed due to their inherent ability to learn the mapping between inputs and outputs by optimizing model parameters to minimize prediction

loss. This capability enables ML models to adapt to the underlying characteristics of a dataset, making them effective tools for investigating historical patterns and predicting future values. Several popular ML methods applied to air quality forecasting belong to the boosting family, including Gradient Boosting, XGBoost [15], LightGBM [11], and CatBoost [9].

Deep learning represents a more advanced and sophisticated class of machine learning models. Although often regarded as "black-box" models due to their limited interpretability, their remarkable ability to capture complex, nonlinear relationships between input features and target variables has made them a dominant approach in forecasting research. Long Short-Term Memory (LSTM) networks are particularly well known for their capability to preserve long-term sequential dependencies, making them a common choice for temporal modeling [13]. A notable extension of LSTM is the Bidirectional Long Short-Term Memory (Bi-LSTM), which processes sequential information in both forward and backward directions to capture richer temporal representations [?]. To improve computational efficiency while maintaining comparable performance, the Gated Recurrent Unit (GRU) has also been widely adopted as a more compact recurrent architecture [4].

Transformer are a powerful candidate due to full data access or selective data access by its different variants, contrasting limited reach and vanishing gradient issues in other deep learning backbones. Hu et al. [3] adopted a Transformer encoder-decoder as the primary temporal forecasting backbone, while spatial dependencies among neighboring grids were modeled through a cross-optimal interpolation layer, where distance-aware learnable weights were computed using a Softmax-like formulation.

To further leverage the complementary strengths of different learning algorithms, ensemble methods have also been extensively explored. By combining multiple predictive models, ensemble approaches can often achieve improved robustness and forecasting accuracy compared with individual models [10, 8].

Hybrid models appear to be a feasible strength due to the diverse roles accompanied by contributing components and thus allow for multi-purpose models. Deep learning seems to join this very much. For instance, Milli et al., combined Prophet to capture long term evolution while afterward LSTM to learn non linear dependencies. Transformers and deep learning models are good combo. Dong et al., [2] utilized empirical mode decomposition (EMD) to simplified data into multiscale fracture, meanwhile self attention learn data relationship and Bi-LSTM for forecast output. Similar approaches have been developed by Liu et al., [7]. Another expressive model is developed by Wei et al., [16] where CNN, LSTM and Attention is stacked in hierarchical architecture and further enhanced by hyperparameter optimization by Whale optimization algorithm (WOA). Another interesting combination by Inam et al., [5] which utilize a simpler form of deep learning is the combination of Polynomial Regression (PR) and Fully Connected Neural Networks (FCNN), which has showed surpassed performance compared to hybrid model of Random Forest (RF) and deep learning and relative better performance compared to hybrid deep learning with LSTM families going regression and boosting families.

Transformers is also a potential candidate for model construction. Zhang and Awang [19] adopted the Inverted Transformer architecture to overcome the limitation of the vanilla Transformer, which embeds all features at a single time step into one token. Instead, the Inverted Transformer treats each feature as an independent token across the entire time span, thereby emphasizing temporal dependencies throughout the sequence rather than inter-feature correlations at individual time

steps.

In conventional Transformer-based forecasting models, multiple input variables are first projected into a common latent embedding space before the attention mechanism is applied. Whether the model represents the data using temporal tokens, feature tokens, or a hybrid strategy, heterogeneous variables such as $\text{PM}_{2.5}$ concentration, wind speed, wind direction, temperature, and humidity are processed through the same sequence of learned transformations. Consequently, the influence of each environmental factor is learned implicitly from data, making it difficult to explicitly incorporate established physical knowledge regarding atmospheric pollutant transport.

However, atmospheric variables possess fundamentally different physical characteristics. Geographical distance constrains the spatial extent of pollutant transport, terrain elevation influences atmospheric dispersion, wind direction determines transport pathways, wind speed governs advective displacement, and the $\text{PM}_{2.5}$ concentration at neighboring stations determines the amount of pollutant available for transport. Treating all of these variables identically may overlook valuable domain knowledge that is already well understood from atmospheric physics.

To address this limitation, this study proposes a **Physics-Aware Spatial Attention (PASA)** mechanism. Instead of relying solely on learned latent similarity, PASA explicitly models the physical effects of geographical distance, terrain difference, wind alignment, advected transport distance, and source $\text{PM}_{2.5}$ concentration as interpretable bias terms. These physics-guided biases are independently weighted by learnable parameters and are directly added to the attention logits before SoftMax normalization. Consequently, the attention mechanism jointly captures both learned feature similarity and physically meaningful transport behavior, while preserving the contribution of each physical factor in an interpretable and independently optimizable manner.

3 Methodology

3.1 Physics-Aware Spatial Attention (PASA)

Let i denote a neighboring (source) station and j denote the target station. Throughout this section, all pairwise quantities are indexed according to the ordered pair (i, j) , representing the potential transport of pollutants from station i to station j . Since multiple neighboring stations may influence the same target station, multiple values of i may correspond to the same target station j .

3.1.1 Station Graph Construction

Each monitoring station has a fixed geographical coordinate consisting of latitude and longitude. To accurately estimate the distance between stations over a large geographical region, the Haversine formula is adopted instead of Euclidean or Manhattan distance. Unlike planar distance measures, the Haversine distance accounts for the Earth’s curvature and therefore provides a more accurate estimate of great-circle distance between two stations. The pairwise distance between stations i

and j is computed as

$$a_{ij} = \sin^2 \left(\frac{\Delta\phi_{ij}}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\Delta\lambda_{ij}}{2} \right), \quad (1)$$

$$d_{ij} = 2R_{\oplus} \arcsin \left(\sqrt{a_{ij}} \right), \quad (2)$$

where $R_{\oplus} = 6371$ km is the Earth's radius, ϕ and λ denote latitude and longitude in radians, respectively, and

$$\Delta\phi_{ij} = \phi_i - \phi_j, \quad \Delta\lambda_{ij} = \lambda_i - \lambda_j.$$

For numerical stability, the implementation clips a_{ij} to the interval $[0, 1]$ before applying the square root and inverse sine functions. Fig. 1 illustrates the Haversine distance on a spherical surface.

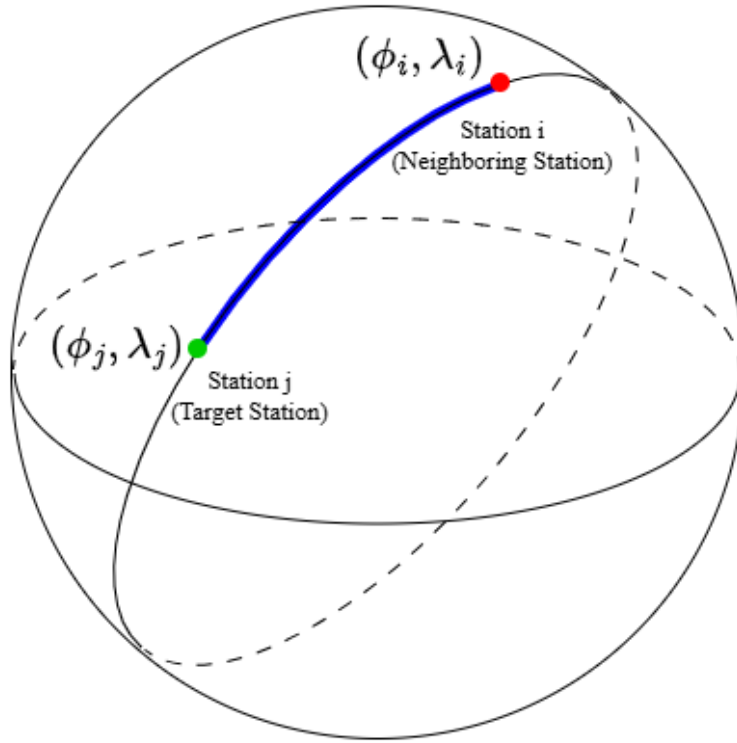


Figure 1: Geometric illustration of the Haversine distance, representing the great-circle distance between two points on a spherical surface.

Although the pairwise distance describes the spatial separation between two stations, it does not indicate the direction of pollutant transport. Therefore, the geographic bearing from neighboring station i to target station j is also computed. The bearing represents the clockwise angle measured from True North to the line connecting station i to station j , and is expressed in radians as

$$\beta_{i \rightarrow j} = \text{atan2}(\sin(\Delta\lambda_{ji}) \cos(\phi_j), \cos(\phi_i) \sin(\phi_j) - \sin(\phi_i) \cos(\phi_j) \cos(\Delta\lambda_{ji})), \quad (3)$$

where

$$\Delta\lambda_{ji} = \lambda_j - \lambda_i.$$

Since station locations remain fixed throughout training, both the distance matrix and the bearing matrix are precomputed once and stored as static buffers.

3.1.2 Neighbor Graph

PM_{2.5} transport is generally localized over a limited spatial range. Therefore, only stations located within a predefined distance threshold are considered neighbors. In this study, the neighborhood radius is fixed at 300 km, forming a static graph throughout training.

The neighborhood of target station j is defined as

$$\mathcal{N}(j) = \{i \mid d_{ij} \leq 300 \text{ km}\}. \quad (4)$$

Consequently, only neighboring stations within a 300-km radius centered at the target station are considered during spatial attention, while all remaining station pairs are masked.

3.1.3 Distance Bias

Besides the learned content similarity, a physics-inspired distance bias is introduced to encourage stronger interactions between geographically closer stations. The distance bias is defined as

$$b_{ij}^{\text{dist}} = \exp\left(-\frac{d_{ij}}{\sigma_d}\right), \quad (5)$$

where $\sigma_d = 200$ km controls the decay rate. Consequently, nearby stations receive larger bias values, whereas the contribution gradually decreases with increasing geographical distance.

3.1.4 Terrain Bias

Topographical differences also influence pollutant transport. Stations separated by large elevation differences are generally less likely to exchange pollutants due to terrain barriers. Therefore, an elevation-based terrain bias is defined as

$$b_{ij}^{\text{terrain}} = \exp\left(-\frac{|e_i - e_j|}{\sigma_h}\right), \quad (6)$$

where e_i and e_j denote the elevations of stations i and j , respectively, and $\sigma_h = 1200$ controls the attenuation caused by terrain differences.

As the elevation difference increases, the terrain bias decreases, reducing the likelihood of pollutant transport between the two stations.

3.1.5 Wind Alignment Bias

Wind direction plays a major role in the transport of atmospheric pollutants. A neighboring station is expected to have a stronger influence when it lies approximately in the direction of the prevailing wind at the source station, while stations located upwind are much less likely to contribute to pollutant transport.

For every ordered station pair (i, j) , the angular difference between the instantaneous wind direction at neighboring station i and the geographic bearing from station i to station j is computed as

$$\theta_{ij}(t) = \psi_i(t) - \beta_{i \rightarrow j}, \quad (7)$$

where $\psi_i(t)$ denotes the wind direction at station i .

The wind direction is computed using the two-argument inverse tangent function,

$$\psi(t) = \text{atan2}(v_{950}(t), u_{950}(t)), \quad (8)$$

where $u_{950}(t)$ and $v_{950}(t)$ denote the zonal and meridional wind components at the 950 hPa pressure level, respectively. The $\text{atan2}(\cdot)$ function correctly determines the quadrant of the wind vector, producing wind directions over the interval $(-\pi, \pi]$.

The wind-alignment score is then defined as

$$b_{ij}^{\text{wind}}(t) = \max(\cos \theta_{ij}(t), 0). \quad (9)$$

The resulting score satisfies

$$0 \leq b_{ij}^{\text{wind}}(t) \leq 1.$$

When

$$|\theta_{ij}| < 90^\circ,$$

station j lies approximately in the downwind direction of station i , resulting in a positive alignment score. The alignment reaches its maximum value of one when the wind direction exactly matches the bearing from station i to station j .

Conversely, when

$$|\theta_{ij}| \geq 90^\circ,$$

the cosine becomes non-positive and is clamped to zero. Consequently, the wind-alignment term provides no additional evidence that pollutants are transported from station i toward station j . This does not remove the station pair from the attention mechanism; rather, it only removes the wind-related contribution to the final attention score.

The geometric interpretation of Eq. (10) can be understood by considering the line passing through station (i) that is perpendicular to the direction from station (i) to station (j). This line partitions the plane into two half-planes. The half-plane containing the vector $i \rightarrow j$ is referred to as the *downwind half-plane*, while the opposite side is the *upwind half-plane*. Only wind vectors lying within the downwind half-plane satisfy $|\theta_{ij}| < 90^\circ$, resulting in a positive cosine value. In this case, the wind has a positive component in the direction of station (j), making pollutant transport from station (i) to station (j) physically plausible. The alignment score reaches its maximum value when the wind direction exactly coincides with the bearing from station (i) to station (j). Conversely, wind vectors located in the upwind half-plane satisfy $|\theta_{ij}| \geq 90^\circ$, yielding $\cos \theta_{ij} \leq 0$. Such winds either transport pollutants away from the target station or have no component directed toward it, and therefore the wind-alignment term is clamped to zero.

This geometric interpretation is visually summarized in Figure 2.

3.1.6 Advected Distance

In practice, PM_{2.5} transport does not necessarily follow the static shortest distance between neighboring station i and target station j . Instead, pollutant movement is influenced by the wind field, where both wind speed and direction determine the displacement of the transported air parcel.

Therefore, rather than measuring the interaction strength using only the stationary distance d_{ij} , we estimate the effective distance after considering the movement of the air parcel during one timestep.

Let A denote the location of the neighboring station i , and B denote the location of the target station j . After one timestep, the air parcel originating from station i is assumed to move from A to A' according to the local wind condition. The objective is then to estimate the distance between the advected air parcel position A' and the target station B , denoted as d_{ij}^{eff} .

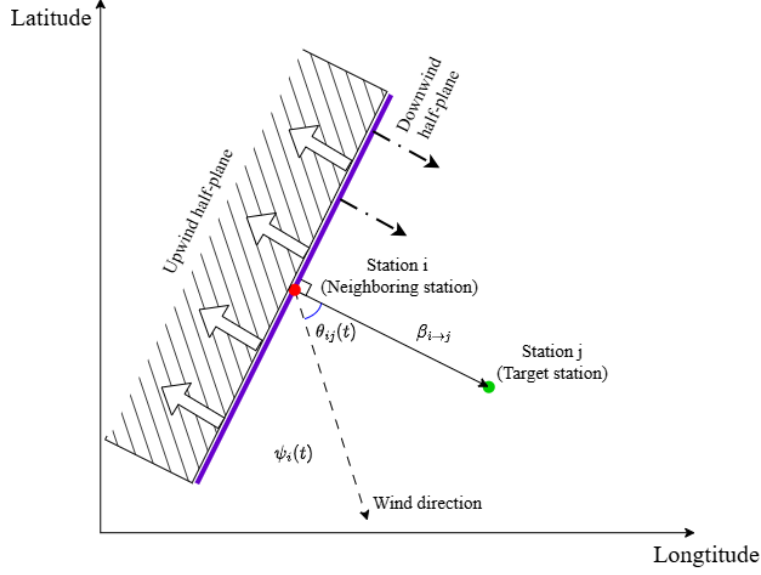


Figure 2: Geometric representation of the wind-alignment mechanism

The displacement of the air parcel is calculated from the wind speed as

$$\Delta_i(t) = u_i(t) \times 3.6 \times \Delta t, \quad (10)$$

where $u_i(t)$ is the wind speed at station i in m/s, 3.6 is the conversion factor from m/s to km/h, and Δt represents the dataset timestep in hours. In this study, $\Delta t = 3$ hours.

The corresponding wind speed is computed as the Euclidean magnitude of the horizontal wind vector,

$$V(t) = \sqrt{u_{950}(t)^2 + v_{950}(t)^2 + \varepsilon}, \quad (11)$$

where $\varepsilon = 10^{-8}$ is a small numerical constant added to improve computational stability. The computed wind direction determines the orientation of pollutant transport.

The original distance between station i and station j is obtained from the Haversine distance described previously:

$$AB = d_{ij}. \quad (12)$$

Furthermore, the angle between the wind direction at station i and the direction from station i to station j is defined as in Eq. (7)

Considering the triangle formed by A , A' , and B , the effective distance $A'B$ can be derived using

the law of cosines:

$$d_{ij}^{\text{eff}}(t) = \sqrt{d_{ij}^2 + \Delta_i(t)^2 - 2d_{ij}\Delta_i(t)\cos(\theta_{ij}(t))}. \quad (13)$$

This formulation allows the spatial relationship between two stations to dynamically change according to the current wind condition. When the wind blows directly from station i toward station j , $\theta_{ij} = 0$, and the effective distance becomes

$$d_{ij}^{\text{eff}} = |d_{ij} - \Delta_i|,$$

indicating that the transported pollutant parcel moves closer to station j . Conversely, when the wind blows away from station j , the effective distance increases, reducing the likelihood of pollutant influence.

Figure 3 illustrates the geometric relationship between a neighboring station and the target station, denoted by points A and B , respectively, where A' represents the advected position of the neighbor after a time interval t .

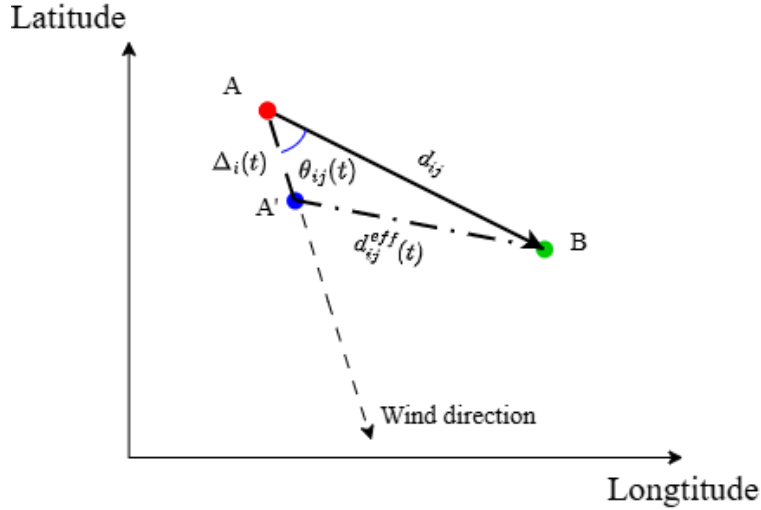


Figure 3: Geometric illustration of a neighboring station (A) and a target station (B), showing the advected position (A') after time t .

Finally, the advected distance is converted into an attention bias using an exponential decay function:

$$b_{ij}^{\text{adv}}(t) = \exp\left(-\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d}\right), \quad (14)$$

where σ_d controls the spatial decay rate.

The advected distance formulation provides a more physically meaningful representation of spatial dependency compared with conventional distance-based graph construction methods. Existing spatio-temporal PM_{2.5} forecasting approaches typically construct spatial relationships using static geographical distance, where the influence between two stations is assumed to remain unchanged regardless of the prevailing atmospheric conditions. Although distance-based graphs effectively capture local spatial correlation, they do not explicitly consider the dynamic movement of pollutants caused by wind-driven transport.

In contrast, the proposed advected distance introduces a dynamic spatial relationship by estimating the relative position of an air parcel after being transported by the current wind field. Therefore, two stations that are geographically distant may still exhibit strong interactions when the wind direction and speed support pollutant transport, while nearby stations may have reduced influence when the wind moves pollutants away from the target location.

By incorporating wind-driven displacement into the spatial attention mechanism, the model can move beyond purely distance-based assumptions and better capture the directional and dynamic nature of atmospheric pollutant propagation. This provides a more realistic approximation of physical transport processes while maintaining the flexibility of data-driven attention mechanisms.

3.1.7 Source Concentration Bias

While the advected distance describes the likelihood that pollutants can be transported from a neighboring station to a target station, it does not account for the amount of pollutant available at the source. In practice, atmospheric transport depends not only on the transport pathway but also on the pollutant concentration carried by the moving air parcel. Consequently, a station with very low PM_{2.5} concentration should contribute less to the target station than a highly polluted station, even if both exhibit similar wind conditions and geographical proximity.

To incorporate this physical intuition, the proposed model introduces a source concentration bias that combines the advected distance with the instantaneous PM_{2.5} concentration measured at the source station.

Let $P_i(t)$ denote the PM_{2.5} concentration at neighboring station i at time t . Since PM_{2.5} concentration is non-negative, the implementation first ensures numerical stability by clipping negative values and adding a small constant

$$P_i^+(t) = \max(P_i(t), 0) + \varepsilon, \quad (15)$$

where $\varepsilon = 10^{-3}$ prevents undefined values when computing the logarithm.

The source concentration bias is then defined as

$$b_{ij}^{\text{mass}}(t) = -\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d} + \log(P_i^+(t)), \quad (16)$$

where $d_{ij}^{\text{eff}}(t)$ is the effective distance derived in the previous subsection and σ_d is the same spatial decay coefficient used in the distance and advected-distance biases.

Using the logarithmic identity

$$\log(a) + \log(b) = \log(ab),$$

the above expression can be interpreted as

$$b_{ij}^{\text{mass}}(t) = \log \left(P_i^+(t) \exp \left(-\frac{d_{ij}^{\text{eff}}(t)}{\sigma_d} \right) \right). \quad (17)$$

Therefore, the bias is proportional to the product of two physically meaningful quantities: the pollutant concentration available at the source station and the transport efficiency determined by the wind-driven effective distance. A neighboring station with a high PM_{2.5} concentration and a favorable transport pathway naturally receives a larger attention bias than a clean station or one located along an unfavorable transport path.

The logarithmic transformation serves two purposes. First, it compresses the dynamic range of PM_{2.5} measurements, preventing the concentration term from overwhelming the other attention components. Second, it converts the multiplicative relationship between source concentration and transport efficiency into an additive bias that can be directly incorporated into the attention score.

3.1.8 Physical-Aware Spatial Attention

After computing all individual bias terms, including the distance bias, terrain bias, wind-alignment bias, advected-distance bias, and source concentration bias, they are integrated with the conventional Scaled Dot-Product Attention score to produce the final attention logits before normalization.

Given the hidden representation h_i of station i , three linear projections are first applied to obtain the Query, Key, and Value vectors. The content similarity between neighboring station i and target station j is first computed as

$$s_{ij}^{\text{content}}(t) = \frac{\mathbf{q}_i(t)^\top \mathbf{k}_j(t)}{\sqrt{d_k}}, \quad (18)$$

where $\mathbf{q}_i$ and $\mathbf{k}_j$ denote the query and key vectors, respectively.

The final attention score is then obtained by

$$s_{ij}(t) = s_{ij}^{\text{content}}(t) + w_{\text{dist}} b_{ij}^{\text{dist}} + w_{\text{terrain}} b_{ij}^{\text{terrain}}(t) + w_{\text{wind}} b_{ij}^{\text{wind}} + w_{\text{adv}} b_{ij}^{\text{adv}}(t) + w_{\text{mass}} b_{ij}^{\text{mass}}(t), \quad (19)$$

where each coefficient w is a learnable scalar parameter optimized jointly with the remaining network parameters. Rather than prescribing the relative importance of each physical factor, the model automatically learns how much each bias contributes to the attention mechanism.

Therefore, the proposed attention mechanism extends the conventional Scaled Dot-Product Attention by incorporating five complementary physics-guided bias terms, enabling the attention weights to jointly reflect latent feature similarity, geographical proximity, terrain effects, wind-driven transport, and the pollutant availability at neighboring stations.

Since the SoftMax function is defined as

$$\alpha_{ij}(t) = \frac{\exp(s_{ij}(t))}{\sum_{k \in \mathcal{N}(j)} \exp(s_{kj}(t))},$$

each attention weight is normalized by the sum of the exponentiated attention scores over all neighboring stations of the target station j . Consequently,

$$\sum_{i \in \mathcal{N}(j)} \alpha_{ij}(t) = 1,$$

meaning that the attention weights form a probability distribution over the neighboring stations. As a result, neighboring stations with larger attention scores contribute larger weights to the spatial context representation, whereas stations with smaller attention scores are assigned proportionally lower weights.

The spatial context vector in Eq. (19) is then obtained as a weighted aggregation of the value vectors from all neighboring stations. Here, the value vector $\mathbf{v}_i(t)$ represents the learned latent representation of neighboring station i , while the attention weight $\alpha_{ij}(t)$ determines the extent to which the information from station i contributes to the representation of the target station j . As a result, neighboring stations that are geographically close, favorably aligned with the prevailing wind, possess a short advected transport path, and exhibit high $\text{PM}_{2.5}$ concentration naturally exert a greater influence on the target station through larger attention weights.

Unlike conventional spatial attention mechanisms that derive attention solely from learned feature similarity, the proposed Physics-Aware Spatial Attention (PASA) explicitly incorporates physically interpretable bias terms into the attention logits before SoftMax normalization. This design allows atmospheric transport knowledge to directly guide the information aggregation process while preserving the adaptive representation learning capability of the Transformer. Furthermore, the learnable coefficients associated with each bias enable the network to automatically determine the

relative importance of individual physical factors during training rather than relying on manually specified weights.

$$\mathbf{c}_j(t) = \sum_{i \in \mathcal{N}(j)} \alpha_{ij}(t) \mathbf{v}_i(t), \quad (20)$$

Therefore, the hidden representation of the target station is updated by selectively aggregating information from all neighboring stations, where the contribution of each neighbor is determined jointly by learned feature similarity and physically motivated transport priors.

3.2 Advective PM_{2.5} Transport Feature

Although the proposed spatial attention mechanism incorporates several physically-inspired bias terms, the attention operation itself is performed on latent hidden representations rather than directly on the raw PM_{2.5} observations. Since each hidden state jointly encodes historical PM_{2.5} concentrations together with multiple meteorological variables, the contribution of neighboring PM_{2.5} sources is learned implicitly during feature extraction. However, from the perspective of atmospheric transport, the amount of pollutant available at neighboring stations is itself an important physical indicator of the pollutant likely to reach the target station.

To explicitly provide this information to the temporal encoder, an additional feature termed the *Advective PM_{2.5} Transport Feature* is introduced. Unlike the spatial attention module, which exchanges learned hidden representations after encoding, this feature is computed directly from the raw PM_{2.5} observations and is concatenated with the original input features before entering the GRU encoder. Consequently, the GRU receives an explicit estimate of the PM_{2.5} transported from surrounding stations while still learning higher-level latent representations through the attention mechanism.

The computation follows the same physical assumptions introduced in the Advective Spatial Attention module. First, the effective transport distance between neighboring station i and target station j is computed using the wind-driven displacement model,

$$d_{ij}^{\text{eff}}(t) = \sqrt{d_{ij}^2 + \Delta_i(t)^2 - 2d_{ij}\Delta_i(t)\cos\theta_{ij}(t)}, \quad (21)$$

where d_{ij} is the Haversine distance, $\Delta_i(t)$ is the wind-driven displacement over one timestep, and $\theta_{ij}(t)$ is the angle between the wind direction at station i and the bearing from station i to station j .

Instead of directly using the effective distance, the transport efficiency is modeled using two physically motivated components. The first component describes attenuation during transport,

$$\omega_{ij}^{\text{path}}(t) = \exp\left(-\frac{d_{ij}^{\text{eff}}(t)}{L_{\text{decay}}}\right), \quad (22)$$

where L_{decay} is a learnable decay length controlling how rapidly pollutant influence decreases with transport distance.

The second component models atmospheric dilution caused by stronger winds,

$$\omega_i^{\text{dilution}}(t) = \frac{1}{1 + \kappa u_i(t)}, \quad (23)$$

where $u_i(t)$ denotes the wind speed at the neighboring station and κ is a learnable parameter controlling the dilution rate. Higher wind speeds increase atmospheric mixing and dispersion, thereby reducing the pollutant concentration arriving at the target station.

The overall transport coefficient is obtained by multiplying the two components,

$$\omega_{ij}(t) = \omega_{ij}^{\text{path}}(t) \omega_i^{\text{dilution}}(t), \quad (24)$$

while stations outside the predefined neighborhood radius are assigned zero transport weight.

Finally, the transported PM_{2.5} feature for target station j is computed as the normalized weighted average of neighboring PM_{2.5} concentrations,

$$T_j(t) = \frac{\sum_{i \in \mathcal{N}(j)} \omega_{ij}(t) P_i(t)}{\sum_{i \in \mathcal{N}(j)} \omega_{ij}(t)}, \quad (25)$$

where $P_i(t)$ denotes the observed PM_{2.5} concentration at neighboring station i . The normalization prevents stations with a larger number of neighbors from producing disproportionately large transport estimates, making the feature comparable across the monitoring network.

Compared with the spatial attention mechanism, which determines how encoded hidden representations are exchanged among stations, the proposed Advective PM_{2.5} Transport Feature provides the encoder with an explicit, physics-guided estimate of incoming pollutant concentration before feature extraction begins. Consequently, the model benefits from both an explicit transport-related input feature and a learned spatial interaction mechanism, allowing the influence of neighboring PM_{2.5} sources to be modeled at two complementary stages of the forecasting pipeline.

3.3 Gated Recurrent Unit (GRU)

Gated Recurrent Unit (GRU) is a recurrent neural network architecture proposed as a simplified alternative to the Long Short-Term Memory (LSTM) network. While both models are designed to address the vanishing-gradient problem encountered by conventional Recurrent Neural Networks (RNNs), GRU achieves comparable performance using a more compact architecture. Unlike LSTM, which employs separate cell and hidden states together with three gating mechanisms, GRU maintains only a single hidden state and utilizes two gates: the *reset gate* and the *update gate*. This simpler design reduces the number of trainable parameters, leading to lower computational cost, faster training, and reduced memory consumption while still preserving the ability to capture long-term temporal dependencies.

The reset gate determines how much historical information should be retained when generating the candidate hidden state. It is defined as

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r), \quad (26)$$

where $\mathbf{x}_t$ denotes the input at time step t , $\mathbf{h}_{t-1}$ is the previous hidden state, $\sigma(\cdot)$ represents the sigmoid activation function, and $\mathbf{W}_r$, $\mathbf{U}_r$, and $\mathbf{b}_r$ are learnable parameters.

Using the reset gate, the candidate hidden state is computed as

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h), \quad (27)$$

where $\odot$ denotes element-wise multiplication. A reset gate value close to zero suppresses historical information, allowing the network to focus primarily on the current input, whereas a value close to one preserves the influence of previous hidden states.

The update gate controls the balance between historical memory and newly generated information. It is computed as

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z). \quad (28)$$

Finally, the hidden state is updated according to

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \quad (29)$$

The update gate adaptively determines how much information is inherited from the previous hidden state and how much is replaced by the candidate hidden state. Consequently, GRU is capable of preserving long-term temporal dependencies while remaining computationally efficient.

3.3.1 Proposed Framework

The complete forecasting framework first receives the historical $\text{PM}_{2.5}$ observations together with the corresponding meteorological variables. In addition, the proposed Advective $\text{PM}_{2.5}$ Transport Feature described in Subsection 3.2 is concatenated to the original input as an additional feature channel. Unlike conventional handcrafted features, this transport feature provides a physically motivated estimate of the amount of $\text{PM}_{2.5}$ advected from neighboring stations toward each target station according to the current wind field. Consequently, the encoder is supplied not only with the local observations but also with an explicit estimate of incoming pollutant transport.

Let the augmented input at time step t be

$$\mathbf{x}'_t = [\mathbf{x}_t, \mathbf{T}_t],$$

where $\mathbf{x}_t$ denotes the original meteorological and $\text{PM}_{2.5}$ observations, and $\mathbf{T}_t$ represents the proposed advective transport feature.

The augmented sequence is then processed by the GRU encoder,

$$\mathbf{h}_t = \text{GRU}(\mathbf{x}'_t, \mathbf{h}_{t-1}),$$

allowing temporal dependencies to be learned through the update and reset gating mechanisms.

After temporal encoding, the proposed Physics-Aware Spatial Attention (PASA) is applied to capture spatial interactions among monitoring stations. Instead of relying solely on the conventional Scaled Dot-Product Attention, PASA incorporates geographically meaningful physical biases—including distance, terrain, wind alignment, advected distance, and source concentration—into the attention score before SoftMax normalization. The resulting attention weights are used to aggregate neighboring hidden representations,

$$\mathbf{c}_j = \sum_{i \in \mathcal{N}(j)} \alpha_{ij} \mathbf{v}_i,$$

where $\mathbf{c}_j$ denotes the spatial context representation for target station j .

Finally, the temporal representation produced by the GRU and the spatial context obtained from PASA are fused through a learnable gating layer,

$$\mathbf{h}_j^{\text{mix}} = \phi([\mathbf{h}_j; \mathbf{c}_j]),$$

where $[\dots; \dots]$ denotes feature concatenation and $\phi(\cdot)$ represents the learnable linear transformation implemented by the fusion layer.

This fusion enables the model to jointly exploit temporal dynamics and spatial dependencies. While the GRU captures the temporal evolution of $\text{PM}_{2.5}$ at each individual station, PASA selectively aggregates information from physically relevant neighboring stations. Consequently, the resulting hidden representation simultaneously models local temporal behavior and regional pollutant transport, providing a more informative latent representation for subsequent probabilistic encoding and forecasting.

A key highlight of the proposed architecture is the integration of physical knowledge at two complementary stages of the learning process. Before temporal encoding, the Advective $\text{PM}_{2.5}$ Transport Feature provides the GRU with an explicit estimate of wind-driven pollutant transport, while after temporal encoding, the proposed Physics-Aware Spatial Attention (PASA) incorporates geographical distance, terrain, wind alignment, advected distance, and source concentration directly into the attention scores. This dual-stage design allows physical knowledge to guide both feature representation learning and spatial information aggregation, resulting in more physically interpretable spatio-temporal modeling.

The Variational Autoencoder (VAE) extends the deterministic autoencoder by introducing a probabilistic latent representation, which is particularly beneficial for modeling the uncertainty and complex variations in spatio-temporal $\text{PM}_{2.5}$ dynamics. Instead of encoding an input feature representation $\mathbf{x}$ into a fixed latent vector, the encoder learns the parameters of a latent distribution, represented by the mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}^2$:

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)).$$

The sampled latent representation $\mathbf{z}$ is then passed to the decoder for reconstruction or prediction. This stochastic encoding allows the model to capture multiple possible representations of pollutant evolution caused by complex interactions between historical concentrations, meteorological conditions, and spatial dependencies.

To regularize the latent space, the Kullback–Leibler (KL) divergence is added to the training objective:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2N} \sum_{n=1}^N \sum_{i=1}^d (1 + \log \sigma_{n,i}^2 - \mu_{n,i}^2 - \sigma_{n,i}^2), \quad (30)$$

which constrains the learned latent distributions to remain close to a standard Gaussian prior. This produces a smooth and structured latent space, improving the model’s ability to generalize under varying spatio-temporal pollution patterns and uncertain future $\text{PM}_{2.5}$ conditions.

4 Experimental Design

4.1 Dataset Description

We evaluate the proposed model on the **KnowAir** dataset [1], a large-scale benchmark for air quality forecasting introduced by PM2.5-GNN [14]. The dataset has subsequently been adopted by numerous state-of-the-art studies, including Air-DualODE [12], AirDDE [17], and EGCFC [18], enabling fair comparisons across different forecasting models.

KnowAir covers a broad region of 184 nodes in China spanning approximately 103°E–122°E and 28°N–42°N, encompassing several heavily polluted urban clusters. Air quality observations are collected from monitoring stations provided by the Ministry of Ecology and Environment (MEE), while meteorological information is obtained from the ERA5 reanalysis dataset released by the European Centre for Medium-Range Weather Forecasts (ECMWF). The dataset records measurements every three hours from January 1, 2015 to December 31, 2018 (GMT).

Following previous studies, PM_{2.5} concentration is the prediction target. Each sample combines historical pollutant observations with multiple meteorological variables describing atmospheric conditions that influence pollutant transport, accumulation, and dispersion. The meteorological variables included in KnowAir are summarized in Table 1.

Table 1: Meteorological variables included in the KnowAir dataset.

Category	Variables
Wind	100m U-component, 100m V-component, 950 hPa U-component, 950 hPa V-component
Temperature	2m temperature, 2m dewpoint temperature, 925 hPa temperature, 950 hPa temperature
Humidity	Relative humidity (950 hPa, 975 hPa), Specific humidity (950 hPa)
Atmospheric Conditions	Surface pressure, Boundary layer height, Vertical velocity (950 hPa), Vorticity (950 hPa), K-index
Precipitation	Total precipitation

The complete dataset spans four years, from January 2015 to December 2018. Following the official benchmark settings, three experimental protocols are provided for evaluating forecasting performance under different temporal scenarios. Protocol 1 uses continuous yearly splits, while Protocols 2 and 3 focus on seasonal and short-term forecasting, respectively. The data partitions are summarized in Table 2.

Table 2: Official data splits of the KnowAir dataset.

Protocol	Training	Validation	Testing	Frequency (hours)
1	2015/01 – 2016/12	2017/01 – 2017/12	2018/01 – 2018/12	3
2	2015/11 – 2016/02	2016/11 – 2017/02	2017/11 – 2018/02	3
3	2016/09 – 2016/11	2016/12	2017/01	3

4.2 Evaluation Metrics

To quantitatively evaluate the prediction performance of each model, two widely used regression error metrics are employed: the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE). The formulations of these metrics are given as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (31)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (32)$$

where N denotes the total number of observations, y_i is the ground-truth PM_{2.5} concentration of the i -th sample, and $\hat{y}_i$ is the corresponding predicted value.

4.3 Configuration and Benchmark Models

The experimental configuration used throughout this study is summarized in Table 3. To ensure a fair and objective comparison among different models while mitigating the effect of random weight initialization and stochastic optimization, each experiment was independently repeated five times using different random seeds. The reported performance is the average of these five runs.

Table 3: Experimental configuration.

Hyperparameter	Value
Batch size	32
Epochs	50
Learning rate	5×10^{-4}
Weight decay	5×10^{-4}
History length	24
Prediction length	24
Early stopping patience	10
Repeated runs	5

The dataset has a temporal granularity of three hours. By taking the historical timestep of 24, the model forecasts the next 24 steps, meaning that the past three days of records are used to estimate air quality conditions over the following three days. This setup is considered sufficient to provide adequate historical information while also maintaining a practical and realistic forecasting horizon for various application scenarios. This configuration is adopted as the standard benchmark setting among the evaluated models.

All experiments were conducted on a workstation equipped with an NVIDIA Tesla T4 GPU with 16 GB GDDR6 memory. The models were implemented using Python 3.13 and PyTorch 2.6.0 with

CUDA 12.4 and cuDNN 9.1 acceleration. All experiments were performed using a single GPU.

5 Result and Discussion

6 Ablation Studies

An ablation study provides a more comprehensive understanding of the proposed model by systematically evaluating the impact of individual components through their inclusion or removal. This analysis reveals the contribution of each component to the overall model performance, highlighting its effectiveness and importance within the complete architecture. Furthermore, the ablation study also investigates the effects of different hyperparameter settings, such as the lengths of the historical input sequence and the forecasting horizon. Although longer historical sequences are generally expected to provide more informative temporal patterns and improve prediction accuracy, this assumption does not always hold in practice. Excessively long input sequences may introduce redundant or noisy information, potentially degrading model performance. Therefore, evaluating different parameter configurations not only provides deeper insights into the model’s behavior but also helps identify an optimal experimental setup that balances predictive accuracy and computational efficiency. In general, three categories of ablation studies are conducted: (1) the effects of different input and output sequence lengths, (2) comparisons of alternative temporal modeling architectures, and (3) component ablation through the removal of individual modules. These experiments provide a comprehensive evaluation of the proposed model by examining the influence of sequence configuration, temporal feature extraction, and each architectural component on the overall forecasting performance.

6.1 Input/Output sequence length

Input length and output length are important experimental configurations, as they determine the amount of historical context available to the model and the complexity of the forecasting task, respectively. It is generally expected that a longer historical window provides richer temporal information and therefore improves model performance, whereas a longer prediction horizon increases forecasting difficulty and may degrade predictive accuracy. In the standard configuration, both the history length and the prediction length are set to 24 time steps, corresponding to 3 days. To investigate the impact of these configurations, two experimental setups are designed. In the first setup, the history length is fixed at 24 time steps, while the prediction length varies among ([8, 16, 24, 48, 72, 96, 120]). In the second setup, the prediction length is fixed at 24 time steps, while the history length varies over the same range: ([8, 16, 24, 48, 72, 96, 120]). Since the dataset has a sampling frequency of 3 hours, these settings correspond to temporal windows of 1, 2, 3, 6, 9, 12, and 15 days, respectively.

6.2 Temporal Modeling Architectures

The core backbone for sequential modeling in the proposed framework is the Gated Recurrent Unit (GRU), which is intentionally selected for its ability to capture temporal dependencies while maintaining relatively low computational cost. Although GRU provides an effective balance between forecasting accuracy and efficiency, numerous alternative architectures have also been specifically developed for time series forecasting, each offering different trade-offs between predictive performance and computational resource consumption. Therefore, evaluating alternative temporal modeling architectures is worthwhile.

To assess the influence of the temporal backbone, the GRU component is replaced with three widely adopted architectures: a Multi-Layer Perceptron (MLP), a Long Short-Term Memory (LSTM) network, and a Temporal Convolutional Network (TCN). These models represent distinct approaches to temporal feature extraction and enable a comprehensive comparison of their forecasting performance within the proposed framework.

7 Conclusion

Despite explicitly considering the influence of $\text{PM}_{2.5}$ concentrations at surrounding stations on the target station across multiple time steps, the proposed model does not directly capture the temporal evolution of particle accumulation and dispersion; instead, neighboring concentrations are incorporated only as additional input features. Future work could address this limitation by introducing a dedicated module to characterize the accumulation, growth, and dispersion of $\text{PM}_{2.5}$ over time, thereby providing a more physically meaningful representation of temporal pollutant dynamics.

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